# Solution Architecture

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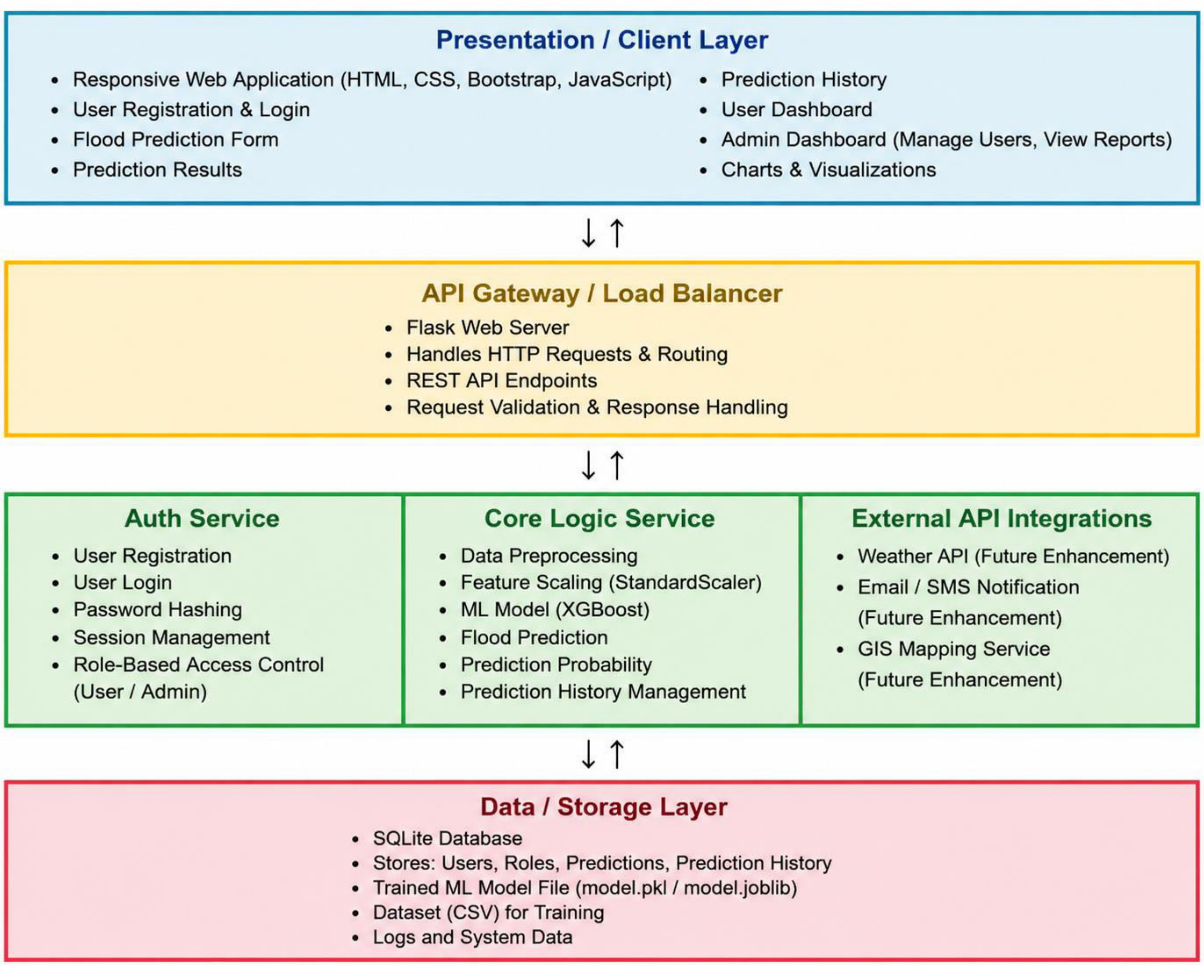
| **Date** | 27 June 2026 |
| **Team ID** | |
| **Project Name** | Rising Waters: A Machine Learning Approach to Flood Prediction |
| **Maximum Marks** | 5 Marks |

**Solution Architecture Diagram:**

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



## Component Description Table

| Component Name | Description / Role in Architecture | Technologies Used |
| --- | --- | --- |
| Presentation Layer | Provides the user interface for registration, login, flood prediction, dashboard, prediction history, and administrator operations. It enables users to interact with the application through a responsive web interface. | HTML5, CSS3, Bootstrap, JavaScript |
| API Gateway / Backend | Receives requests from the client, performs authentication, validates input data, invokes the machine learning model, communicates with the database, and returns prediction results. | Flask, Python |
| **Core Logic Service** (Microservices) | Handles data preprocessing, feature scaling, machine learning prediction, prediction history management, and business logic. It processes user inputs and generates flood prediction results. | XGBoost, Scikit-learn, Pandas, NumPy, Joblib |
| Database | Stores user information, authentication details, prediction history, trained model references, and application data required for system operation. | SQLite |